

. Which one of the following statements concerning lanthanides elements is false

- (a) Lanthanides are separated from one another by ion exchange methods
- (b) The ionic radii of trivalent lanthanides steadily increase with increase in atomic number**
- (c) All lanthanides are highly dense metals
- (d) Most typical oxidation of lanthanides is +3

Answer: (b) The ionic radii of trivalent lanthanides steadily increase with increase in atomic number

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

102. Complex ion is shown by

- (a) Ag
- (b) Au
- (c) Cu
- (d) All of these**

Answer: (d) All of these

Explanation: Transition-metal ions form complexes readily because of their relatively high charge density and available orbitals for accepting ligand electron pairs.

103. Which of the following transition metal is used as a catalyst

- (a) and
- (b)
- (c) Gold
- (d) Both**

Answer: (d) Both

Explanation: Transition metals are effective catalysts because they can change oxidation state and form intermediate complexes/adsorb reactants on their surfaces.

104. For which of the following transition metal ion is shown highest magnetic moment having outer electronic configuration

- (a) 3d¹
- (b) 3d⁸
- (c) 3d⁵**
- (d) 3d⁷

Answer: (c) 3d⁵

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d⁰ and d¹⁰ species are diamagnetic.

105. The number of unpaired electrons is maximum in (Atomic no. : Ti = 22; V = 23; Cr = 24; Fe = 26)

- (a) Cr**
- (b) Fe



- (c) V
(d) Sc

Answer: (a) Cr

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

106. Europium is

- (a) s-block element
(b) p-block element
(c) d-block element
(d) f-block element

Answer: (d) f-block element

Explanation: The correct choice is f-block element. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

107. Which of the following elements is alloyed with copper to form brass

- (a) Lead
(b) Silver
(c) Zinc
(d) Antimony

Answer: (c) Zinc

Explanation: The correct choice is Zinc. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

108. In which of the following metallic bond is strongest

- (a) Fe
(b) Sc
(c) V
(d) Cr

Answer: (d) Cr

Explanation: The correct choice is Cr. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

109. Which of the following is a colourless ion

- (a) Cu²⁺
(b) Fe³⁺
(c) Ti³⁺
(d) Zn²⁺

Answer: (d) Zn²⁺

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

110. The substance used in cancer therapy is

- (a) Rn
(b) Ni
(c) Fe
(d) Co



Answer: (d) Co

Explanation: The correct choice is Co.

This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

111. In solution of AgNO_3 , if Cu is a solution become blue due to

- (a) Oxidation of Ag
- (b) Oxidation of Cu**
- (c) Reduction of Ag
- (d) Reduction of Cu

Answer: (b) Oxidation of Cu

Explanation: The correct choice is Oxidation of Cu. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

112. Lanthanide for which + II and + III oxidation states are common

- (a) La**
- (b) Nd
- (c) Ce
- (d) Eu

Answer: (a) La

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

113. The number of unpaired electrons in Zn^{++} is

- (a) 2
- (b) 3
- (c) 4
- (d) 0**

Answer: (d) 0

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

114. The first transition element is

- (a) Chromium
- (b) Scandium**
- (c) Nickel
- (d) Copper

Answer: (b) Scandium

Explanation: The correct choice is Scandium. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

115. The electronic configuration (outermost) of Mn^{+2} ion (atomic no. of Mn=25) in its ground state

- (a) $3d^5 4s^0$**
- (b) $3d^4 4s^1$
- (c) $3d^3 4s^2$
- (d) $3d^2 4s^2 4p^2$



Answer: (a) $3d^5 4s^0$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

116. The highest magnetic moment is shown by the transition metal ion with the outer electronic configuration

- (a) $3d^2$
- (b) $3d^5$**
- (c) $3d^7$
- (d) $3d^9$

Answer: (b) $3d^5$

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d^0 and d^{10} species are diamagnetic.

117. Which has valency two

- (a) Al
- (b) Rb
- (c) Cu**
- (d) Ge

Answer: (c) Cu

Explanation: The correct choice is Cu. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

118. All the following statements about the transitional elements are true except that

- (a) All of the transitional elements are predominantly metallic
- (b) In aqueous solution many of their simple ions are coloured
- (c) Most of the transitional elements show pronounced catalytic activity
- (d) Most of the transitional elements show only one valence state**

Answer: (d) Most of the transitional elements show only one valence state

Explanation: The correct choice is Most of the transitional elements show only one valence state. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

119. Which of the following transition metals can have highest oxidation state

- (a) Cr
- (b) Co
- (c) Ni**
- (d) Cu

Answer: (c) Ni

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.



120. Which of the following melts in boiling water

- (a) Gun metal
- (b) Wood's metal**
- (c) Monel metal
- (d) Bell metal
- (e) None of these

Answer: (b) Wood's metal

Explanation: Melting/boiling trends depend largely on the strength of metallic bonding, which is influenced by the number of valence d and s electrons available for bonding.

121. On the basis of position in the electrochemical series, the metal which does not displace hydrogen from water and acid is

- (a) Hg**
- (b) Al
- (c) Pb
- (d) Ba

Answer: (a) Hg

Explanation: Use the electrochemical series and the stability of the oxidation states involved to compare oxidising/reducing strength or reactivity.

